

Impact of GDP per Capita on Vehicle Registrations using Time Series Tools

Internship Report 2025

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1 Abstract

This internship report investigates the dynamic relationship between macroeconomic indicators and vehicle demand in India over a 25-year period (2000–2024) using time series analysis. The study uses data from Vahan Sewa, the International Monetary Fund (IMF) and the World Bank to evaluate how Nominal GDP, Real GDP, and GDP per capita influence overall vehicle registrations, two-wheeler and four-wheeler.

Methodologically, the data was standardized using Z-scores and detrended via linear regression residuals to achieve stationarity, as verified by Augmented Dickey-Fuller (ADF) tests. Although only the GDP per Capita was successfully made stationary. Granger Causality testing revealed that GDP per capita possesses significant predictive power for overall vehicle registrations specifically at short-term horizons (1-year and 2-year lags), though its influence fades by the third year. Separately, GDP per capita was found to Granger-cause two-wheeler registrations at lags 1 and 2, but only predicted four-wheeler registrations at lag 2. Vector Auto Regression (VAR) models at an optimal lag of 2 were subsequently deployed to forecast registration residuals for the next 10 years.

Beyond statistical modeling, the report explores consumer preferences and socioeconomic factors impacting transit choices. Despite two-wheelers accounting for a staggering 44% of Indian road accident deaths in 2022—with a disproportionately higher fatality rate in rural areas due to inadequate infrastructure—the demand for two-wheelers continuously outpaces four-wheelers. An income elasticity of demand of 1.35 for two-wheelers compared to 1.06 for four-wheelers, demonstrate that two-wheelers remain highly responsive to income growth. Under-developed roads, overwhelming congestions everywhere during busy hours, corrupt infrastructures play important roles in driving these preferences. The study concludes that although income is a good predictor of vehicle ownership, in a developing country such as ours, other factors also play a crucial role in determining vehicle ownership patterns.

2 Introduction

This internship was undertaken at the Indian Statistical Institute, Kolkata (Dunlop campus), in an offline mode. The core focus of the project was Time Series Analysis; a crucial statistical tool widely applied in modelling and interpreting data observed over time. Its importance lies in enabling analysts to extract underlying patterns, make forecasts, and support informed decision making across diverse domains, including economics, finance, and policy planning.

In the context of national development, vehicle ownership serves as a significant indicator of both economic prosperity and infrastructural progress. Whether in advanced economies such as the United States or Japan, or in developing nations like India, the increase of vehicles, especially private two-wheelers and four-wheelers, reflects improvements in individual income levels, mobility, and access to infrastructure.

Simultaneously, Gross Domestic Product (GDP) and GDP per capita are widely recognized as benchmark indicators of economic health and average citizen welfare. However, the question arises: Can GDP or GDP per capita serve as reliable predictors of vehicle ownership in a country like India?

To explore this, the present study employs different time series techniques on total vehicle registrations, GDP, and GDP per capita in India over a 25-year period (2000–2024). Special attention is given to understanding the category-wise patterns of two-wheeler and four-wheeler registrations, examining their respective income elasticities, and identifying consumer preferences. The broad objective is to uncover the dynamic relationships among these economic variables and derive insights that can inform policy and planning in the transportation and infrastructure sectors.

3 Definitions & Preliminaries

Nominal GDP: The total market value of all final goods and services produced in a country in a given period, measured at current prices (not adjusted for inflation).

Real GDP: The total market value of all final goods and services produced in a country, adjusted for inflation (using constant prices from a base year). This allows for meaningful comparisons over time. GDP adjusted for inflation using constant prices from a base year is given by:

$$Real\ GDP = \frac{Nominal\ GDP}{GDP\ Deflator} \times 100$$

GDP per Capita: A measure of economic output per person, calculated by dividing GDP(nominal) by the country's population. It indicates average economic well-being but does not reflect income inequality.

Time Series Components: A time series consists of

Trend(T_t): Long-term growth direction

Seasonality(S_t): Regular periodic fluctuations

Cyclic(C_t): Irregular long-term cycles

Random component(R_t): Irregular or unpredictable noise

Additive Model: $Y_t = T_t + S_t + C_t + R_t$

Multiplicative model: $Y_t = T_t * S_t * C_t * R_t$

Stationarity: A time series is said to be stationary if its statistical properties do not change over time.

ADF Test: Checks for a unit root (non-stationarity) in a time series.

- Null hypothesis H_0 : unit root exists.
- Reject H_0 if test statistic < critical value (series is stationary).

Granger Causality: Tests whether past values of a variable X help predict another variable Y (beyond just past values of Y). If yes, X "Granger-causes" Y .

$$Y_t = \alpha + \sum_{i=1}^p \beta_i Y_{t-i} + \sum_{i=1}^p \gamma_i X_{t-i} + \epsilon_t$$

Y_t = current value of Y , α = intercept, β_i = coefficients of lagged values of Y , γ_i = coefficients of lagged values of X , ϵ_i = errors, p = number of lags.

- Null hypothesis H_0 : $\gamma_i = 0$ for all i (no causation).
- Reject H_0 if test statistic < critical value if X 's lags are statistically significant (eg.via F -tests).

AIC (Akaike Information Criterion): Estimates model quality by balancing fit and complexity (lower AIC = better). AIC prefers complexity so it generally prefers higher lags but contains risk of overfitting.

BIC (Bayesian Information Criterion): Estimates model quality by balancing fit and complexity (lower BIC = better). So, it generally prefers lower lags.

VAR (Vector Auto Regression) Model: Multivariate time series model where each variable is a linear function of its own lags and lags of other variables. Typically used to predict a certain time series with the help of another time series.

Income Elasticity of Demand: The percentage change in vehicle demand given a 1% change in GDP per capita.

4 Objectives

The primary objectives of this summer internship project are as follows:

- To analyse the long-term time series data on total vehicle registrations, nominal GDP, and GDP per capita in India over a 25-year period (2000-2024), in order to understand underlying trends and economic patterns.
- To employ advanced time series modelling techniques, such as Vector Auto Regression (VAR) and Granger Causality tests, to investigate the directional relationship between economic growth (captured through GDP and GDP per capita) and vehicle demand in India.
- To evaluate the predictive power of GDP per capita as a determinant of overall vehicle registrations, assessing its robustness and limitations in explaining demand fluctuations.
- To investigate consumer preferences by comparing the growth patterns and income responsiveness of two-wheeler versus four-wheeler registrations, and to identify key socio-economic and infrastructural factors influencing these preferences.
- To estimate and compare the income elasticity of demand for two-wheeler and four-wheeler ownership, based on changes in GDP per capita, thereby quantifying the responsiveness of each category to income growth.
- To analyse road accident and vehicle ownership data in rural and urban regions, with the aim of deriving meaningful conclusions regarding accessibility, safety, and economic disparities in transportation choices.

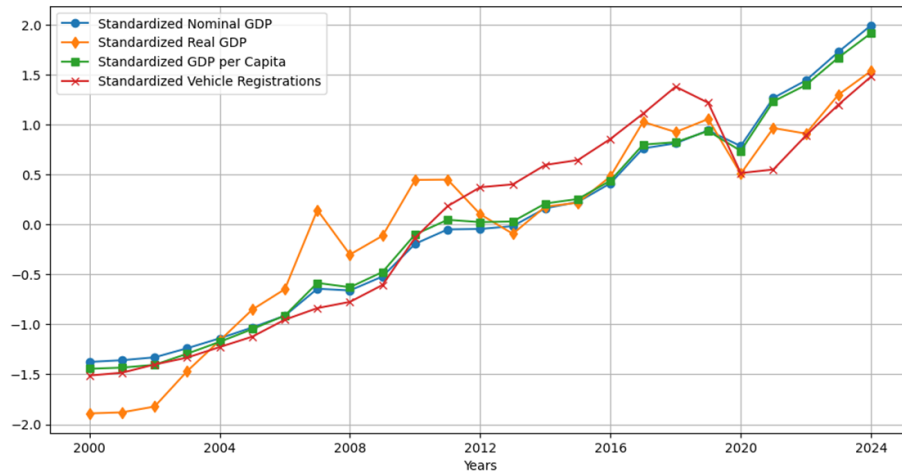
5 Work Done

5.1 Data Collection, Organization & Initial Observations:

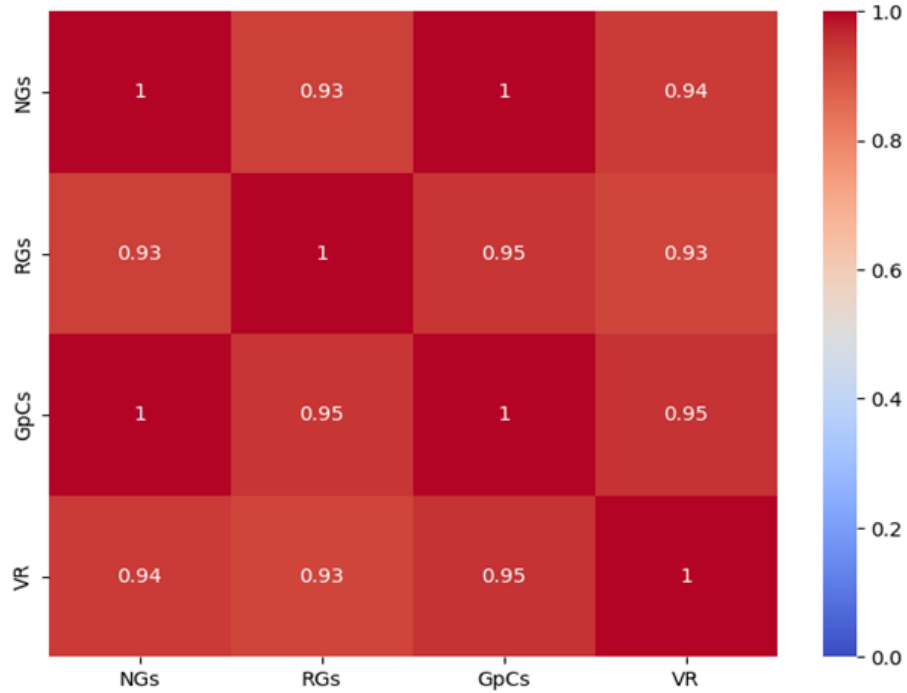
This project work, which was done using Excel and Python programming, started with an extensive data collection. The total yearly vehicle registration data along with the 4-wheeler and 2-wheeler registration data 2000-2024 were all collected from the government website called Vahan Sewa (vahan.parivahan.gov.in). For the Nominal GDP and GDP per capita I obtained those data from IMF (imf.org). To obtain the Real GDP, another metric called GDP deflator was need which was also obtained from world bank (data.worldbank.org). The following formula was used:

$$Real\ GDP = \frac{Nominal\ GDP}{GDP\ Deflator} \times 100$$

After obtaining, the data was organized in Excel and the Real GDP was calculated. In python, mainly Pandas and Seaborn were used. The first one provides an easy-to-use and high performance data structures and methods for data manipulation. The latter is built on top of matplotlib and provides a high-level interface for drawing attractive statistical graphics [1]. Then, the data was then plotted against the years. Though the graph was not functional as the metrics had vastly different units (e.g. Vehicle registration in thousands and GDP in Billions USD). To get rid of the problem and standardize the data, Z-score standardization was used, which works as follows: $x_s = \frac{x - \mu}{\sigma}$ Where x_s = standardized value, x = original value, μ = mean of the data set, σ = standard deviation of the data set. Following is the post-standardization graph:



A correlation heat map was made of these four metrics which can be seen below:



NGs: standardized Nominal GDP

RGs: standardized Real

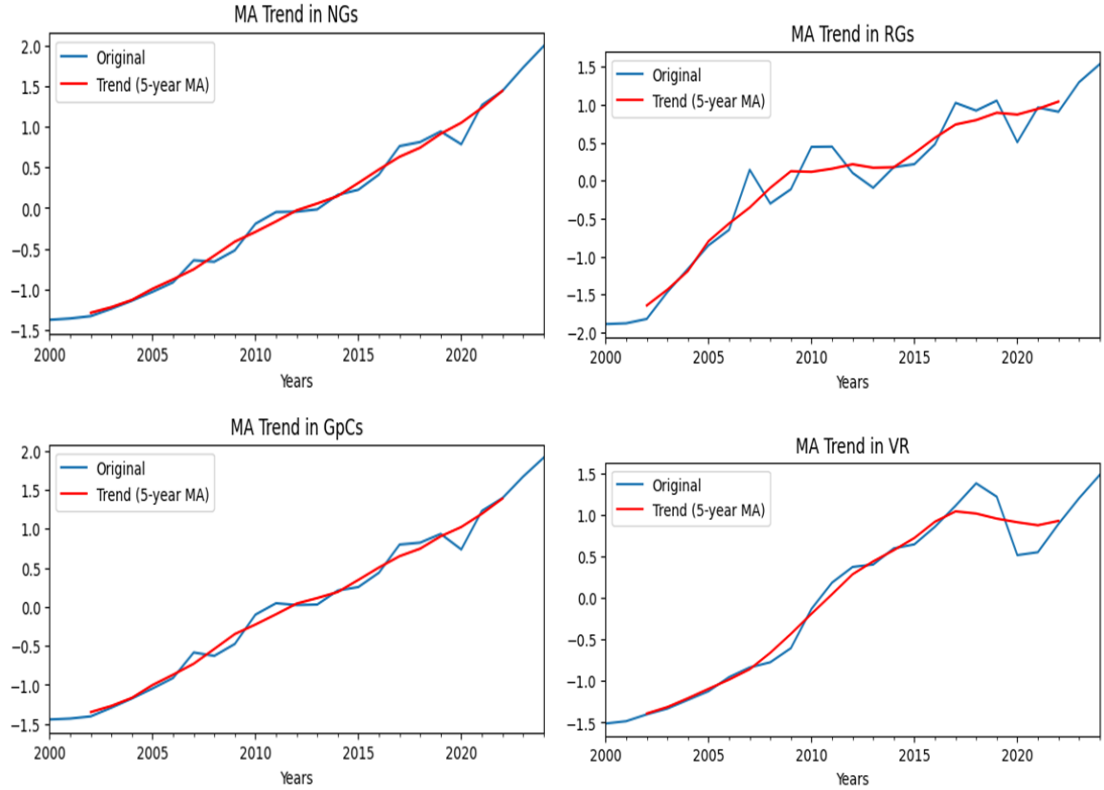
GpCs: standardized GDP per capita

VR: standardized Vehicle Registrations

Our dependent variable is the Vehicle Registrations and we can see high positive correlations between this and the other three metrics. However, it is rather obvious considering all the series are increasing steadily. Now, as the famous saying goes, “Correlation does not necessarily imply Causation”. So, our next step is checking the causality using the Granger Causality test. But since it assumes stationarity, checking the data sets for stationarity shall be the necessary.

5.2 Stationarity Checking & De-trending:

First let us plot the 5-year MA (moving average) to the four time series data:



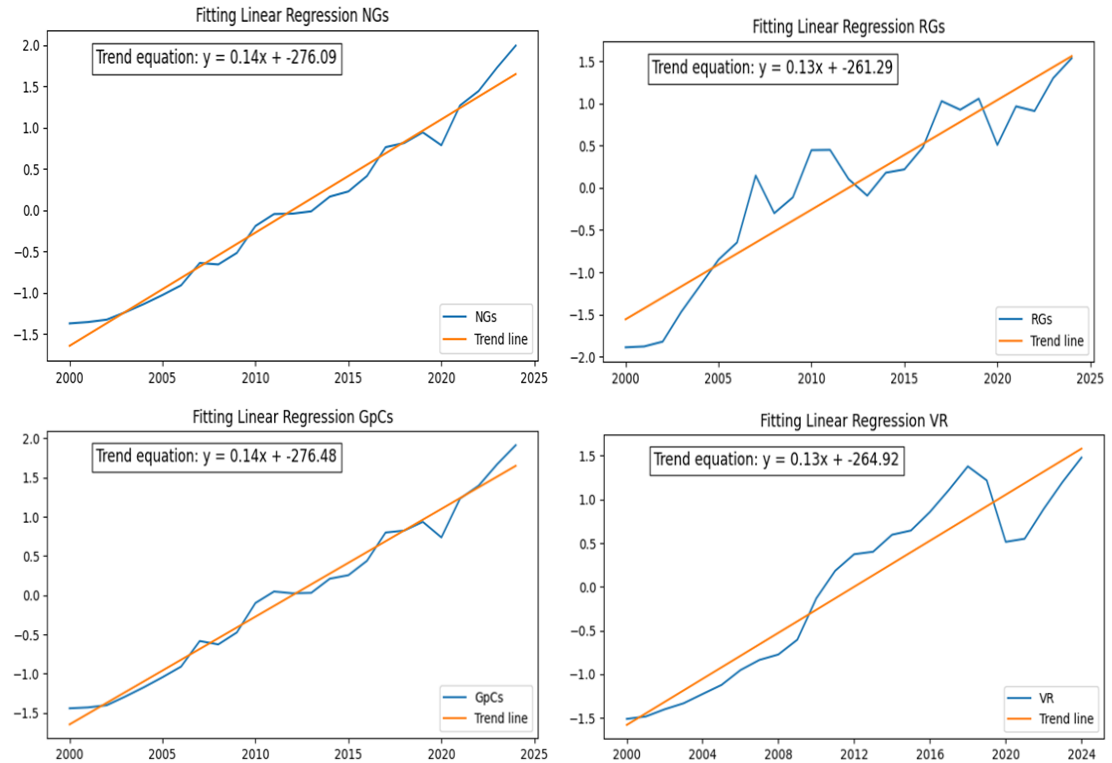
It is quite clear that the moving average has an upward trend, which means the data is not stationary. However, to thoroughly test stationarity ADF (Augmented Dickey-Fuller) test was used. It checks if there is a unit root in the time series data, which essentially means that a shock (abrupt change) affects it permanently i.e. it cannot revert back over time. Now, if p value ≥ 0.05 , then we reject null hypotheses and say that it is stationary. Following is the table consisting of time series vs their p value.

Table 1: p value table before de-trending

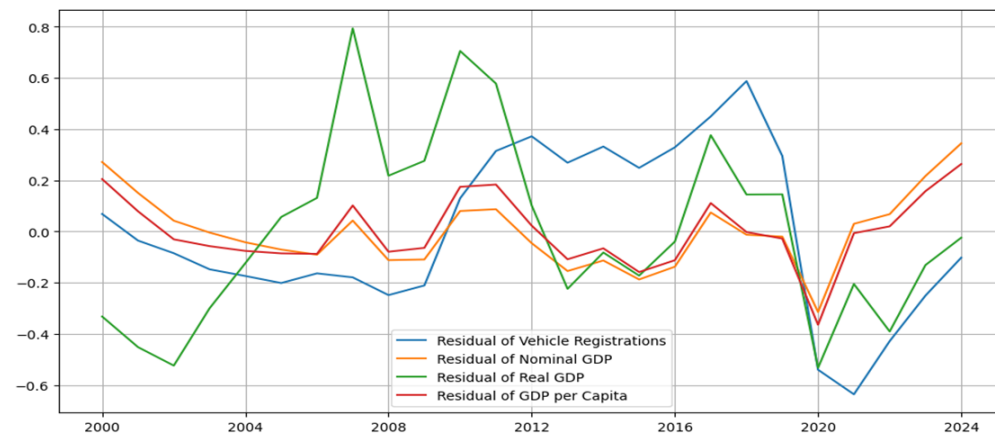
NGs	0.996
RGs	0.670
GpCs	0.991
VR	0.361

So, we can safely conclude that they are non-stationary.

Now, to make them into stationary, we shall be fitting a linear trend line and subtract it to get the residuals, essentially de-trending them. Following are the plots after fitting linear regression:



Thus, the residuals of each time series are obtained which looks as follows when plotted against years:



Now, again ADF test shall be used to see if de-trending made the data stationary. Following is the p value table. Here, RNGs= residual of NGs, RRGs=

residual of RGs, RGpCs= residual of GpCs & RVR= residual of VR:

Table 2: p value table after de-trending

RNGs	0.11
RRGs	0.130
RGpCs	0.018
RVR	0.000009

From this table we claim that de-trending successfully made the GDP per capita and Vehicle Registration data stationary, but Nominal and Real GDP remained non-stationary. So, we shall now check if GDP per capita is of any help in predicting Vehicle Registrations.

5.3 Causality checking & Forecasting:

We shall test the causality using Granger Causality test, which tests if a time series can contribute in predicting another time series in each lag. Thus, it will be used to see if having past values of GDP per capita along with past values of Vehicle Registrations improve the AR model for a certain lag. If it does improve, we say GDP per Capita Granger Causes Vehicle Registrations. Here, if p value < 0.05 , we reject H_0 : GDP per capita does not Granger Cause Vehicle Registrations.

Below is the lag vs p value table:

Table 3: Lag vs p value table

Lags	p-values
1	0.0496
2	0.0146
3	0.0934
4	0.2051
5	0.1258
6	0.2110
7	0.4649

It can be seen that p value is > 0.05 for lags 1 and 2 only, meaning GDP per capita has significant predictive power for Vehicle registrations at 1-year and 2-year lags but slowly fades by year 3, suggesting it alone is insufficient for long-term forecasts. There may be other data that influence Vehicle registrations like fuel prices and government policies.

Now, in order to forecast Vehicle registrations, we shall use VAR (Vector Auto Regression) model. Unlike regular univariate AR model, VAR allows variables to depend on its own past values as well as past values of other variables.

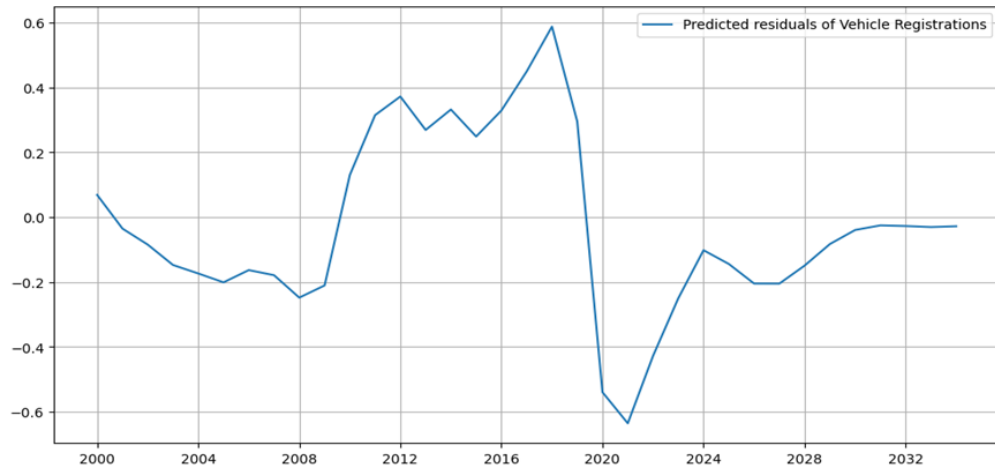
To use this model, the optimal lag is first to be determined, which shall be done on the basis of AIC and BIC. We take the lag for which, both is minimum (*shows minimum).

Table 4: AIC, BIC and Lags

Lags	AIC	BIC
0	-5.718	-5.619
1	-7.251	-6.955
2	-7.452	-6.957*
3	-7.108	-6.416
4	-6.979	-6.089
5	-7.338	-6.250
6	-7.436	-6.150
7	-7.707*	-6.223

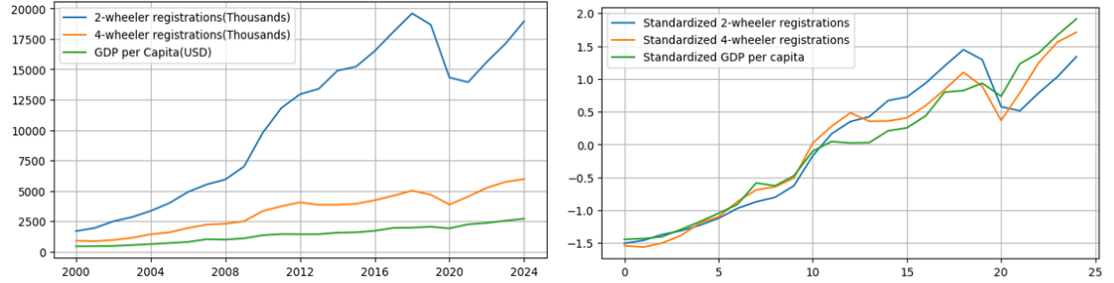
We see that AIC prefers lag 7 while BIC prefers lag 2. However, AIC is second least for lag 2 so that shall be used.

Using VAR model at lag 2, forecasting the residuals of Vehicle Registrations for the next 10 years was done:



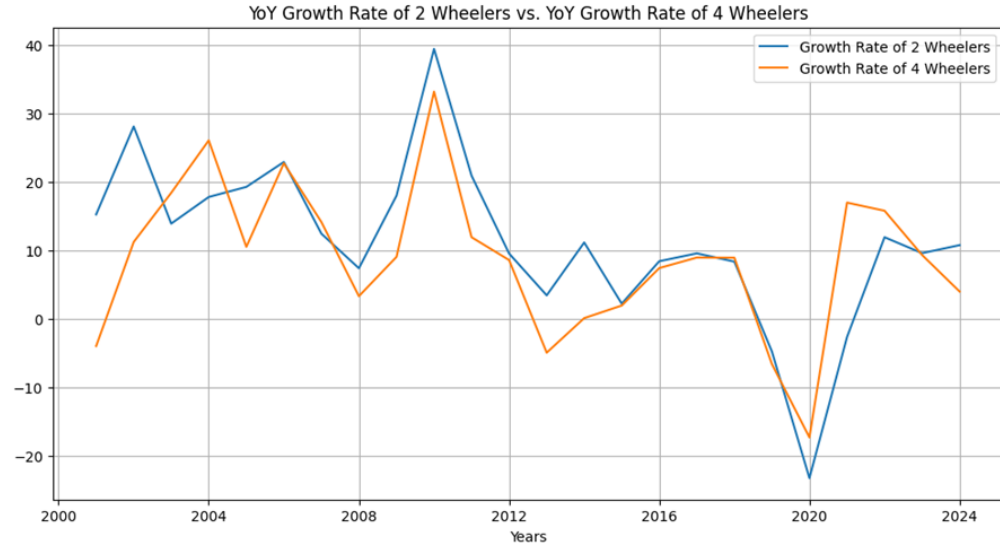
5.4 About 4-Wheelers & 2-Wheelers:

Apart from analyzing the impact of GDP per Capita on Vehicle registrations, its impact on 4-wheelers and 2-wheelers were also analyzed. Finding significant insights about the preference of people between them and the probable causes were also some of the main aims of this project. Below are pre-standardization and post-standardization plots:



Recent reports agree with the findings that 2-wheelers dominate the domestic Indian auto market, accounting for 75.04% of market shares in FY25 [2]. Now in order to see how people's tendency to buy 2-wheeler and 4-wheeler have changed throughout the years, the Year-on-Year growth rate (YoY in short) was calculated. It shows how much has the value increased in a year w.r.t the previous year. $YoY \text{ of year } n = \frac{(Value \text{ in year } n) - (Value \text{ in year } n-1)}{Value \text{ in year } n-1} * 100$

The plot is given below:



Some important observations:

- People always tend to buy 2-wheeler more. Though the growth rate of 4-wheeler has on some occasions beat that of the 2-wheelers.

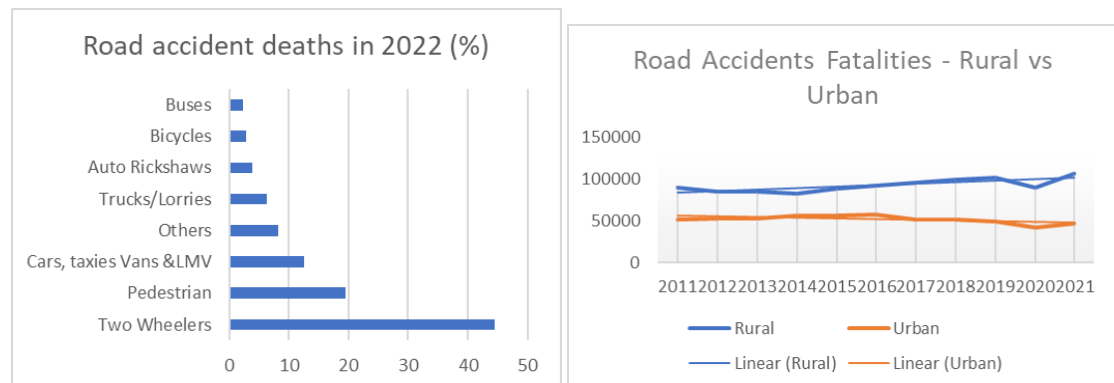
- In 2008 due to the bankruptcy of globally influential investment bank Lehman Brothers, there came an economic recession where we can clearly see both 2-wheeler's and 4-wheeler's growth rate plummeted down. Though right after that, in about 2010, a boom occurred and both of their growth rates shot up to nearly 40% and 35% respectively.

- In the infamous covid year, 2020, both 2-wheelers and 4-wheelers took huge damage falling to -20% and -15% respectively. And in this exceptional case, 2 wheelers exhibited significant decline. Notably in the aftermath, 4-wheeler's growth rate was higher for a few years, likely because of social distancing and isolation protocols.

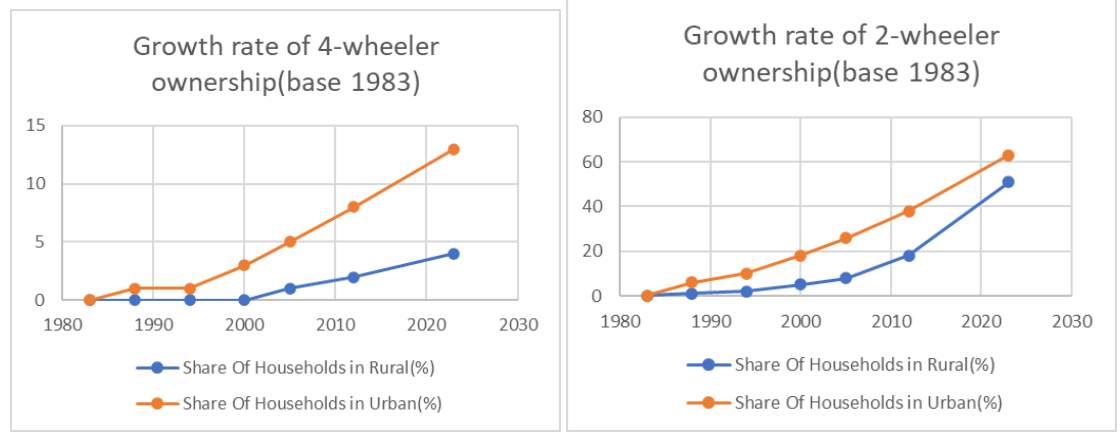
Though, to answer the question about why is 2 wheelers so sought after in India there could be various and numerous parameters. The probable causes include:

- 1) Narrow and under developed roads being not suitable for 4-wheelers.
- 2) India being a highly congested country, traffic jams are more than frequent, which gives 2-wheelers advantage over 4 -wheelers.
- 3) Government policies and subsidies.
- 4) 2-wheelers being more affordable and Indian population being rather economically weak (ranked 122 in GDP per capita ranking 2025, source: World Bank).

Now, let's look into some accident and vehicle ownership data (sources include dataforindia.com, HCES, MoRTH) to see if we can make some meaningful observations.:



In 2022, 2-wheelers accounted for a staggering 44% of all road accident deaths in India. Paradoxically, the demand for 2-wheelers continues to surpass that of 4-wheelers, with the gap steadily widening. As illustrated in the adjacent graph, road accident fatalities are significantly higher in rural areas. This trend has been consistent over the years, with rural areas consistently reporting a higher share of fatalities [3]. This is attributable to inadequate infrastructure, including poor road conditions and limited transportation options. But what can be said about 2-wheeler and 4-wheeler ownership?



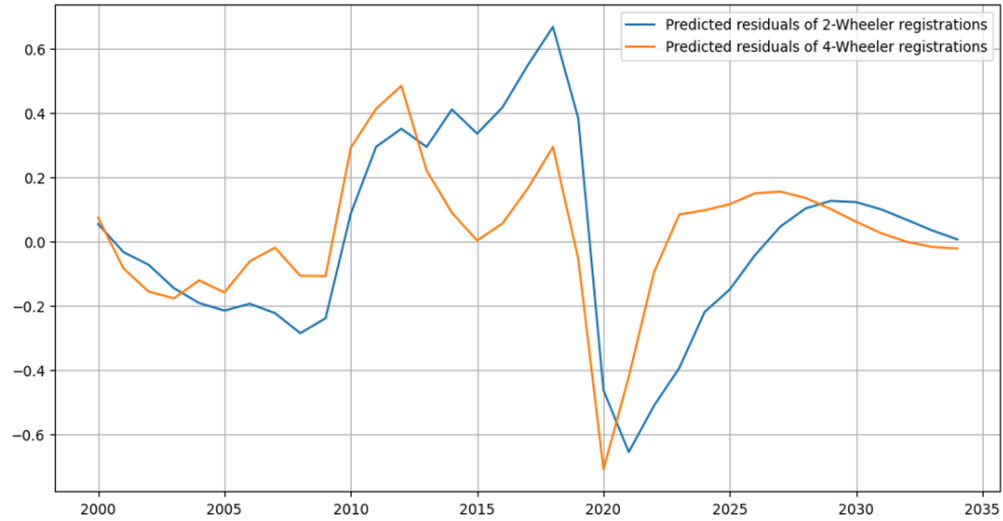
From the graphs shown above it is clear that although urban households lead, by far, in owning both 2 and 4-wheelers, rural households almost exclusively own the former, except maybe some wealthy farmers.

So, it can be established that even though 2-wheelers face highest fatalities in road accidents, the rural population (which is majority) can only resort to them, as rural roads are definitely not eligible for cars.

Now, to check the importance of affordability, the **Income Elasticity** is calculated for 2 and 4-wheelers. It is done by running log-log regression in python and calculating the coefficient of the independent variable, GDP per capita, to check which one is more responsive to it. For 2-wheelers it is 1.35 whereas for 4-wheelers it is 1.06 meaning, for 1% change in GDP per capita there is a 1.35% change for 2-wheelers and 1.06% for the 4-wheelers. Implying even if per capita income rose, people would still prefer 2-wheelers over 4-wheelers.

To further investigate the underlying relationships, the same tests and models were conducted on 2 and 4-wheeler vehicle registrations separately. After standardizing the metrics, Linear Regression model was fitted, residuals were obtained. Residuals were confirmed to be stationary using ADF test (p-values: 0.00007 for 2-wheelers, 0.0009 for 4-wheelers, and 0.018 for GDP per capita). Granger Causality test revealed GDP per capita to be a good predictor of 2-wheelers for lag 1 and 2, but it is only a good predictor of 4-wheelers for lag 2.

Lastly, the VAR model was used to predict the residuals of the registrations of the 2 and 4-wheeler vehicles for lag 2 for the next 10 years. Following is the result:



6 Key Learnings:

- Collecting, cleaning, organising and handling real world data.
- Looking at a data from different angles to get important insights and observations.
- Using python to perform time series analysis and forecasting.
- Finding out that albeit various studies has shown, just GDP per capita is insufficient in predicting vehicle demand, at least in our country.
- The preference of the population between 4-wheelers and 2-wheelers and the probable causes for it.

7 Conclusion:

A famous study done in 1999 which worked with vehicle ownership and GDP per capita 1960- 1992 for various countries and showed that GDP per capita can strongly predict vehicle ownership [4]. This study highlights that although income is a good predictor of vehicle ownership, in a developing country such as ours, other factors play a crucial role in determining vehicle ownership patterns.

Also, this study contributes in understanding the preferences of the population regarding vehicle choice and probable causes which influence them. India is a large country, with Nominal GDP expected to hit 4 trillion USD in 2025. Yet under-developed roads, overwhelming congestions everywhere during busy hours, corrupt infrastructures, poor condition of healthcare forces the majority of Indians to resort to the obviously hazardous option of 2-wheelers. These findings have important implications for policymakers, transportation planners, and safety experts seeking to promote safer and more sustainable transportation options in India.

References

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- [2] “Automobiles Industry Report,” 2025
- [3] “Road Accidents in India - 2022,” MoRTH, 2023.
- [4] J Dargay, D Gately, “Income’s effect on car and vehicle ownership, world-wide: 1960-2015,” 1999.